

VTU SEE Jan 2024: Database Management [BCS501] Solved

Visvesvaraya Technological University - Solved Paper

--- VISVESVARAYA TECHNOLOGICAL UNIVERSITY BELAGAVI ---

Fifth Semester B.E. Degree Examination, Jan 2024 - Database Management [BCS501]

Time: 3 Hours | Max. Marks: 100 | Scheme: 2022 CBCS

--- MODULE 1: RELATIONAL ALGEBRA ---

Q1 [a] Explain Entity Integrity and Referential Integrity with examples. [10M]

Q1 [b] Write Relational Algebra queries for Company Schema. [10M]

--- MODULE 2: NORMALIZATION ---

Q2 [a] Prove why every relation in BCNF is also in 3NF, but not vice-versa. [10M]

SOLVED: Proof using functional dependency $X \rightarrow A$ and candidate key analysis.

Q2 [b] Decompose Relation $R[A,B,C,D,E]$ into 3NF preserving dependencies. [10M]

Result: Complete faculty-evaluated solutions with diagrams.